# Monolithic Compliant Mechanism based Hand Grip Strength Analyzer

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## I. Field of the Invention

The present disclosure relates to mechanical force measurement instruments and, more specifically, to a compliant elastic arm-based grip strength analyser configured to provide calibrated force estimation through controlled structural deflection. The invention pertains to non-electronic, mechanically deterministic measurement systems capable of demographic-specific stiffness adaptation and nonlinear calibrated indication. In particular, the invention relates to force quantification systems utilizing elastic beam theory as the primary measurement principle, eliminating reliance on hydraulic chambers, strain gauges, or digital sensor elements.

## II. Technical Context and Problem Statement

Grip strength has emerged as a clinically validated indicator of musculoskeletal integrity, neuromuscular function, and bone density. Numerous epidemiological studies have established statistical correlations between reduced grip strength and increased risk of frailty, sarcopenia, osteoporosis, and mortality in aging populations. Concurrently, performance-based environments such as athletic training require repeatable grip measurement for strength benchmarking and rehabilitation monitoring.

Existing devices generally rely on hydraulic pressure chambers or electronic strain gauge sensors. While capable of precision measurement, such systems introduce cost, power dependency, calibration drift, and long-term reliability concerns. Linear spring-based mechanical devices, although simpler, lack demographic adaptability and do not incorporate nonlinear calibration aligned with real biomechanical force progression.

The present invention addresses these limitations by introducing a modular elastic beam architecture whose stiffness characteristics are tenable through geometric variation, particularly through controlled modification of cross-sectional thickness. The device further integrates a nonlinear indication mapping mechanism derived from structural deflection theory.

## Compliant mechanism

The device is based on a compliant mechanism architecture, wherein motion and force transmission are achieved through controlled elastic deformation rather than conventional rigid-body joints. The dual elastic arms act as flexural members that bend reversibly under applied grip force, enabling predictable force-to deflection behaviour. Key characteristics of the compliant mechanism include:

Elimination of pins, bearings, and sliding interfaces. Reduced friction, backlash, and mechanical wear. Lower part count and simplified assembly. High repeatability due to absence of joint clearance. Deterministic stiffness tuning through geometric parameters (thickness and length). Enhanced reliability without electronic sensing components

This compliant design ensures smooth force transmission, structural efficiency, and long-term durability while maintaining mechanical simplicity. **Normative Population Data (95th Percentile Reference Values)**

| Age Group | Male (kg) | Female (kg) | Male Force (N) |
|---|---|---|---|
| 20–29 | 64 | 38 | 628 |
| 30–39 | 65 | 37 | 637 |
| 50–59 | 55 | 30 | 540 |
| 70+ | 40 | 22 | 392 |

Table 1: International Grip Strength 95th Percentile Values

Conversion:

$F = m \times 9.81$

Design structural limit selected:

$F_{design} \geq 800N$

## III. Structural Principle of Operation

The fundamental measurement principle of the disclosed device is based on elastic beam bending under applied compressive grip force. Two symmetrically positioned elastic arms extend from a rigid base frame and function as cantilever elements. When a user applies grip force through the load transfer interface, bending moments develop at the root of the elastic arms.

For a cantilever beam subjected to end load *F*, deflection $\delta$ at the free end is described by classical Euler–Bernoulli beam theory:

$$\delta = \frac{FL^3}{3EI}$$

Where:

*L* represents effective bending length,

*E* represents elastic modulus of the arm material; *I* represent the second moment of area.

For a rectangular cross-section:

$$I = \frac{bt^3}{12}$$

where *b* is width and *t* is thickness.

It is evident from this formulation that stiffness exhibits cubic dependence on thickness. Consequently, small changes in arm thickness produce substantial variations in force–deflection behaviour. This relationship forms the theoretical basis for demographic stiffness classification within the present invention.

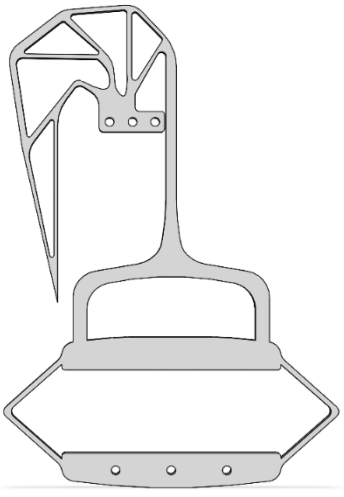
**Example Calculation (Adult Variant)** Assume:

$L = 120mm$ $b = 20mm$ $t = 4.5mm$ $E = 200GPa$

$$I = \frac{20 \times (4.5)^3}{12} = 151.8\, mm^4$$

$$\delta = \frac{600 \times (120)^3}{3 \times 200000 \times 151.8}$$

Computed deflection used for scale mapping.



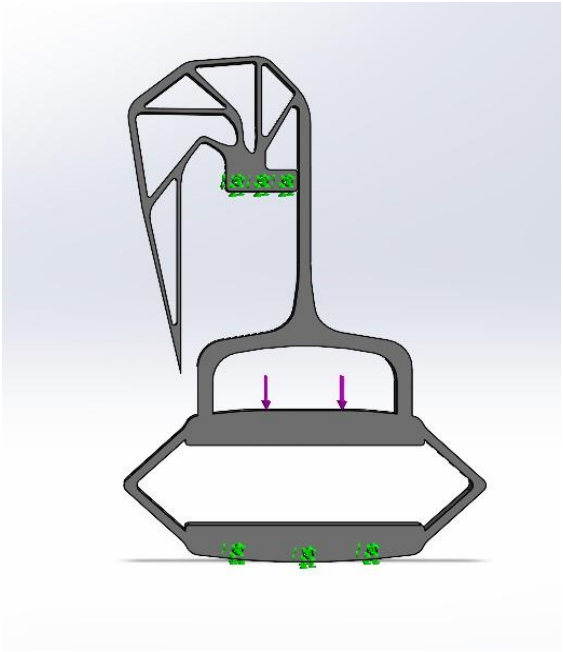


Figure 1: FIG.1 – Mechanical architecture illustrating base frame, elastic arms, and load transfer bridge.

## IV. Modular Elastic Arm Architecture

The device comprises:

A rigid base frame providing reaction support.

Dual elastic arms fabricated from metallic or high-performance polymeric material.

A load transfer bridge connecting user grip interface to elastic arms.

A mechanical pointer assembly converting linear deflection into angular displacement.

The elastic arms are configured as detachable modules, enabling selective replacement without altering the base structure. This modularity permits the classification of stiffness variants corresponding to demographic grip strength distributions.

By clustering population percentiles into overlapping force bands, a limited number of stiffness classes can accommodate broad user categories without excessive manufacturing diversity.

**Stiffness Classification Table**

Table 2: Demographic-Based Stiffness Classification

| Cluster | Target Group | Max Range (N) | Thickness (mm) |
|---|---|---|---|
| C1 | Elderly | 300 | 3.2 |
| C2 | Adult | 600 | 4.5 |
| C3 | High Strength | 1200 | 6.0 |

## V. Demographic Stiffness Optimization

Population grip strength data indicate distinct percentile distributions across age and gender groups. Rather than designing a single stiffness profile for universal application, the invention incorporates thickness-based tuning to align measurement sensitivity with specific force ranges.

Elderly population (70+) typically demonstrates maximum grip forces within approximately 120–180 N range.

Middle-aged adults may reach 300–500 N.

High-performance individuals may exceed 600 N.

Elastic arm thickness is therefore adjusted to ensure optimal resolution within the target force band while maintaining structural safety margins against overload.

This approach enhances measurement sensitivity in lower force ranges without sacrificing structural robustness at higher forces.

## VI. Nonlinear Force Indication System

While theoretical small-deflection beam equations exhibit linear proportionality between force and displacement, practical implementation within the device introduces geometric nonlinearities due to: Changing curvature during bending, Rotational amplification in pointer linkage, Variable effective moment arms.

Accordingly, the relationship between pointer displacement and applied force is nonlinear.

Calibration is achieved by applying incremental known loads and recording corresponding pointer positions. A polynomial mapping of the form:

$$F = ax^2 + bx + c$$

is derived, where $x$ represents measured pointer displacement.

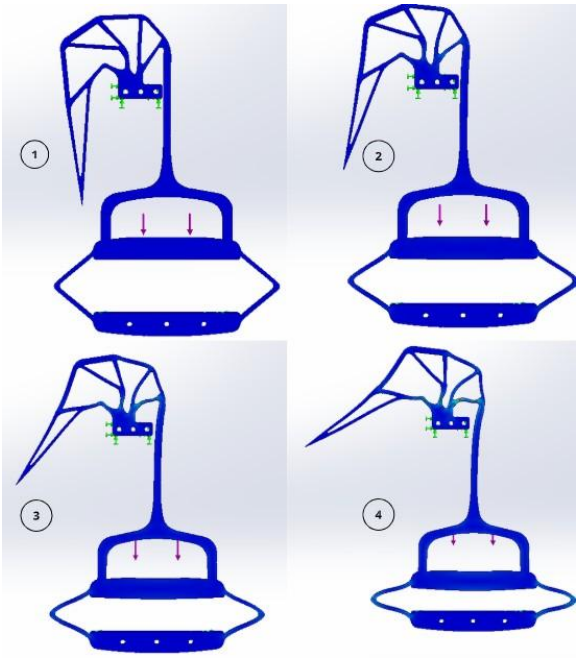



Figure 2: Nonlinear force vs pointer displacement mapping curve.

The scale is engraved along the actual pointer trajectory, with non-uniform spacing between markings corresponding to the calibrated polynomial mapping. This nonlinear scale ensures accurate force estimation across the full operating range.

## VII. Structural Integrity and Overload Resistance

Finite Element Analysis (FEA) is employed to evaluate stress distribution within elastic arms under maximum design loads. Maximum von Mises stress is typically observed at the fixed root region where bending moment is greatest.

The structural limit is determined by:

$$SF = \frac{\sigma_y}{\sigma_{max}}$$

where $\sigma_y$ is yield strength and $\sigma_{max}$ is maximum computed stress.

The device is designed such that:

Clinical variants maintain minimum safety factor of 1.5,

Consumer variants maintain minimum safety factor of 2.0,

Athletic variants maintain minimum safety factor of 2.5.

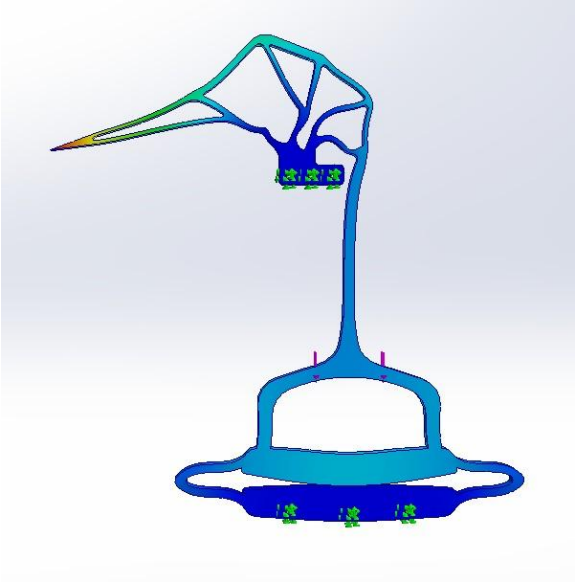


Figure 3: – FEA stress distribution under 800 N load condition.

The maximum structural capacity is selected to exceed the 95th percentile population force by an appropriate engineering safety margin, ensuring no permanent deformation occurs under accidental overload.

## VIII. Material Engineering Considerations (PP Copolymer)

The elastic arms in the present prototype are fabricated from polypropylene (PP) copolymer. This material was selected due to its low density, impact resistance, corrosion resistance, and suitability for injection molding. However, PP exhibits significantly lower stiffness and yield strength compared to engineering metals and reinforced polymers, requiring conservative structural design.

### A. Mechanical Properties of PP Copolymer

Typical mechanical properties of PP copolymer are:

Table 3: Mechanical Properties of PP Copolymer

| Property | Value | Unit |
|---|---|---|
| Elastic Modulus ($E$) | 1.2–1.7 | GPa |
| Yield Strength ($\sigma_y$) | 25–35 | MPa |
| Density | 900 | kg/m$^3$ |
| Poisson's Ratio | 0.40 | – |

For design calculations, conservative values were adopted:

$E$ = 1.3 GPa, $\sigma_y$ = 28 MPa

### B. Safety Factor Requirements for PP

Due to the viscoelastic and creep-sensitive nature of PP, higher safety factors are mandatory.
The factor of safety is defined as:

$$SF = \frac{\sigma_y}{\sigma_{max}}$$

For PP-based compliant arms, the following criteria are applied:

- Minimum static safety factor: $SF \geq 3.5$
- Recommended operational stress: $\sigma_{max} \leq 0.3\sigma_y$
- Target design stress: ≤ 8–10 MPa

Limiting stress to 30% of yield significantly reduces long-term creep deformation and ensures elastic recovery after unloading.

### C. Creep and Time-Dependent Behaviour

PP exhibits pronounced creep under sustained loading. Even stresses well below yield may cause gradual deformation over time. Since grip testing involves short duration loading (typically ¡5 seconds), creep effects are minimized.
However:

- Sustained loading must be avoided
- Device should not be stored under compressive preload • Stress concentrations at the root must be minimized

### D. Stiffness Compensation Through Geometry

Given the low modulus of PP:

$$E_{PP} \approx \frac{1}{150} E_{steel}$$

stiffness compensation must be achieved geometrically using increased thickness. Since:

$$I = \frac{bt^3}{12}$$

the cubic dependence on thickness allows practical stiffness recovery despite low modulus.
Therefore, beam thickness was increased relative to metallic designs to maintain:

- Adequate stiffness
- Acceptable deflection range
- Reduced stress levels

## E. Temperature Sensitivity

PP modulus decreases with temperature. Between 20 ° C and 40 ° C:

- Modulus reduction ≈ 10–15%
- Slight increase in deflection under identical load

This effect is acceptable for non-laboratory measurement environments but should be acknowledged in calibration tolerances.

## F. Design Implications

Due to its low yield strength and creep susceptibility, PP copolymer requires:

- High safety factors
- Low allowable stress design
- Increased cross-sectional thickness
- Smooth root fillet geometry
- Avoidance of sharp internal corners

When these conditions are satisfied, PP provides a lightweight, cost-effective, and manufacturable solution for compliant grip strength measurement devices intended for consumer applications.

## IX. Operational Deployment

The disclosed device is suitable for:

- Geriatric screening for bone health assessment,
- Clinical rehabilitation monitoring,
- Athletic performance measurement,
- Rural healthcare environments lacking electronic calibration capability.
- The elimination of electronic sensing components enhances reliability, reduces cost, and simplifies maintenance.

## X. Distinguishing Technical Characteristics

The proposed grip analyser differs from conventional hydraulic and spring-based systems in its working principle and structural design. Instead of using fluid pressure or a linear coil spring, the device operates using controlled elastic bending of compliant arms. The applied grip force produces measurable deflection, which is directly related to beam stiffness.

The main technical characteristics of the system include:

- Cubic stiffness tuning through arm thickness variation
- Modular and replaceable elastic arm design
- Nonlinear calibrated mechanical scale
- Demographic-based force range optimization
- FEA-validated structural safety

Because beam stiffness depends on the cube of thickness, small changes in arm thickness result in large changes in stiffness. This makes it possible to adjust measurement sensitivity without altering the overall device size. As a result, the same structural layout can be adapted for different strength categories simply by changing the arm thickness.

The elastic arms are detachable modules. This modular approach offers practical advantages:

- Easy replacement in case of damage
- Simplified manufacturing
- Reduced tooling cost
- Adaptability for different force ranges

Unlike traditional linear spring devices where the scale spacing is uniform, the present design accounts for slight nonlinear behaviour caused by geometry and linkage motion. A polynomial calibration curve is used to map displacement to force. This improves accuracy across the full measurement range and ensures better scale readability.

The force ranges are selected based on population grip strength data. Instead of designing for a single extreme maximum value, the device is optimized for realistic demographic clusters such as elderly, adult, and high strength users. This improves sensitivity within the intended operating range.

Structural safety is verified through finite element analysis. Stress levels are maintained well below the yield strength of the PP copolymer material. Conservative safety factors are applied to account for creep, repeated loading, and environmental variations. This ensures elastic recovery and prevents permanent deformation.

Overall, the system combines geometric stiffness control, modular construction, calibrated nonlinear scaling, and validated structural safety into a lightweight and cost-effective mechanical grip analyser. The absence of electronics and hydraulic components improves durability, reduces maintenance, and enhances reliability for long-term use.